

GenStack: Dual-Branch Mixture of Experts for Generalizable Face Forgery Detection

Supplementary Material

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Table S1. Full routing-granularity sweep. All rows except $K=1$ use a predicted multinomial-LR gate on ℓ_2 -normalised CLIP CLS with soft mixing. KMeans clusters are formed by clustering the 23 generators in CLIP feature space. The headline saturates at $K \approx 15$.

K	ID	CM	CF	CD	Avg
1 (global)	94.16	99.20	85.74	78.91	88.68
4 (split gate)	94.85	99.03	92.62	81.14	91.15
10 (KMeans)	94.88	99.14	93.17	84.56	92.33
12 (KMeans)	94.87	99.16	93.36	84.61	92.39
15 (default)	94.95	99.40	94.96	84.53	92.83
18 (KMeans)	94.92	99.43	94.94	84.53	92.82
20 (KMeans)	94.91	99.47	94.98	84.59	92.85
23 (per-gen, GS-23)	94.91	99.43	94.96	84.57	92.83

This supplementary contains additional ablations, per-split breakdowns, and analyses for GENSTACK that were computed but omitted from the main paper due to space constraints. All numbers follow the same 5-fold out-of-fold cross-validation protocol as the main paper. Section letters and table/figure indices are prefixed with “S” to distinguish them from the main paper.

A Full K -Sweep Saturation Curve

The main paper reports five granularity points ($K \in \{1, 4, 10, 15, 23\}$) in Table 3. Table S1 fills in the intermediate points $K \in \{12, 18, 20\}$, all under the same predicted soft-gate protocol. Accuracy rises monotonically up to $K=15$ and then plateaus: $K \in \{15, 18, 20, 23\}$ all lie within 0.03 pp of each other on the headline, confirming that $K=15$ is at the saturation elbow.

B Comparison with Naive Fusion Baselines

A reviewer may ask: “why not simply average the two branches, or use an OR/AND rule?” Table S2 answers this directly. Naive fusions plateau near 88%, only 0.16 pp above Prism alone. The MoE design is required to lift the headline; both branches are required to reach 92.81%.

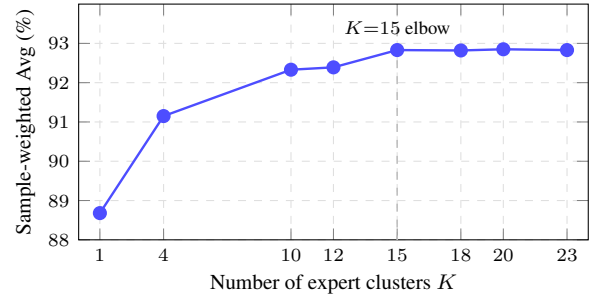


Figure S1. Routing-granularity saturation. Sample-weighted Avg accuracy on HydraFake as a function of the number of expert clusters K . All points except $K=1$ use a predicted soft-mixed gate. Accuracy plateaus at $K \approx 15$; $K \in \{15, 18, 20, 23\}$ are statistically tied within 0.03 pp. We adopt $K=15$ as our default GENSTACK.

C $K=15$ Cluster Composition

Table S3 shows how the 23 HydraFake generators are partitioned into the 15 KMeans clusters used by default GENSTACK. Eleven generators with distinctive forensic fingerprints occupy their own cluster; four multi-generator clusters group semantically related models that share an expert via soft mixing. The clustering matches forensic intuition without using any per-generator labels.

D Soft vs Hard vs Oracle Gate (per-split)

The main paper reports the headline of three gate variants in §4.3. Table S4 provides the per-split breakdown. The soft mixture beats both predicted hard top-1 routing and an oracle gate that knows the true generator label, with the largest gain on cross-forgery (+1.5 pp over hard, +1.2 pp over oracle). Soft mixing recovers samples where the gate’s most-likely cluster is wrong but a related cluster is correct – an option hard one-of- K routing cannot exploit even with perfect labels.

Table S2. MoE vs naive fusion baselines on the full HydraFake test set (52,266 images). Naive averaging only adds +0.16 pp over Prism alone; OR/AND rules underperform Prism. The $K=15$ MoE adds +4.4 pp over simple averaging; removing Prism from the MoE input loses -4.0 pp. Both branches and the MoE design are necessary.

Method	ID	CM	CF	CD	Avg
<i>Single branches</i>					
Pact alone	86.59	98.98	79.52	77.85	84.95
Prism alone	94.90	98.30	84.91	78.08	88.22
<i>Naive fusions (no MoE, no gate)</i>					
Simple avg ($\alpha=0.5$)	94.70	99.10	83.24	79.58	88.38
OR rule	89.55	99.48	87.60	78.72	88.01
AND rule	91.93	97.80	76.83	77.20	85.16
<i>Single-branch MoE ($K=15$ KMeans)</i>					
Pact-only $K=15$ MoE	86.65	98.88	92.92	79.76	88.77
Prism-only $K=15$ MoE	94.90	98.30	84.91	78.08	88.22
GENSTACK $K=15$ (ours)	94.88	99.40	94.96	84.53	92.81

Table S3. $K=15$ cluster composition. How the 23 HydraFake generators are partitioned into the 15 KMeans clusters used by default GENSTACK.

Cluster type	Generators
11 singletons	FaceForensics++, facevid2vid, StyleGAN, ICLight, StarGANv2, DeepFaceLab, Dreamina, FFIW, GPT-4o, Hailuo, InfiniteYou-CD
Consumer (CM)	diffusion AdobeFirefly + StarryAI
Modern (CM/ID)	diffusion Flux1.1Pro + Infinity + HART + Midjourney
Face (mixed)	edit/restore CodeFormer + FaceAdapter + MAGI + Hallo2
ID-preserving face gen (CF)	InfiniteYou-CF + PuLID

E Headroom-vs-Lift per Generator

Table S5 provides the per-generator data behind the headroom-vs-lift analysis in §4.4. Pearson correlation between best-base headroom and GENSTACK lift is $r=0.86$. The table also lists the disagreement rate between Pact and Prism, illustrating that disagreement *rate* alone does not predict lift – DeepFaceLab and FFIW have the highest disagreement (40.7% and 26.0%) but small lifts (+0.36 and +1.16) because the two branches fail in correlated ways on those generators.

F Inference-Time View Masking

We probe how PRISM-SFT relies on each input view by masking views at inference (without retraining) on a stratified subset of 484 test images (~ 22 per generator, real/fake balanced where the test JSON allows). Views 2 and 3 are replaced with a uniform gray 448×448 image of the same size; the prompt is unchanged. Table S7 reports PRISM-SFT-only accuracy under each view configuration. The model’s be-

Table S4. Soft vs Hard vs Oracle gate, per-split. All rows use $K=23$ per-generator experts so that an oracle gate is well-defined. Soft mixing beats the oracle on every split.

Gate	ID	CM	CF	CD	Avg
Predicted hard top-1	94.70	98.50	94.66	84.36	92.45
Oracle (knows true gen.)	94.98	99.70	93.49	84.38	92.50
Predicted soft (ours)	94.91	99.43	94.96	84.57	92.83

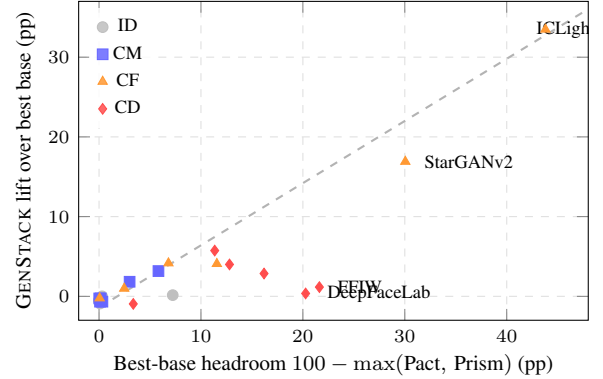


Figure S2. GENSTACK’s lift concentrates where the better base detector has residual headroom. Each point is one of 23 HydraFake generators, coloured by split. Pearson $r=0.86$ between best-base headroom and lift. Saturated generators (bottom-left) have near-zero lift; the largest gains come from headroom-rich generators. DeepFaceLab and FFIW are correlated-error outliers.

haviour under blank views is not directly comparable to a from-scratch retrain on the same view-subset, since PRISM-SFT was trained to expect three informative views and confabulates content for blanks. Numbers are reported here only as a coarse upper bound on each view’s standalone contribution at inference; we leave a clean retrain-on-subset multi-view ablation to future work.

G GENSTACK Robustness on the JPEG/Blur Subset

Figure 3 in the main paper reports robustness for the discriminative branch PACT on the full HydraFake test set. We additionally evaluated the full GENSTACK pipeline (with cached PRISM-SFT verdicts on the same images) on a stratified 1,500-sample subset spanning the seven ID and CM generators for which PRISM-SFT predictions exist under each degradation. Table S8 reports GENSTACK accuracy alongside PACT on the same subset. The MoE preserves graceful degradation: under both JPEG and blur, GENSTACK tracks the PACT curve closely and never amplifies failure. On this subset both numbers are inflated relative to the main-paper full-set numbers because the subset is dominated by easy ID/CM generators where PACT alone already exceeds 94%, and GENSTACK has little headroom to add.

Table S5. Per-generator complementary-error analysis. For each of 23 HydraFake generators we report: number of test samples; PACT accuracy; PRISM-SFT accuracy; GS-15 accuracy; disagreement rate between PACT and PRISM-SFT hard predictions; and the GS-15 lift over the better single base.

Generator	n	PACT	PRISM-SFT	GS-15	Dis.	Lift
<i>Cross-Forgery (CF)</i>						
ICLight	2082	56.20	55.62	89.67	8.4	+ 33.48
StarGANv2	2000	61.15	69.95	86.85	18.9	+ 16.90
InfYou-CF	3244	84.06	93.19	97.35	14.5	+4.16
FaceAdapter	294	88.44	69.05	92.52	21.4	+4.08
PuLID	3360	92.86	97.53	98.51	6.2	+0.98
CodeFormer	1750	92.74	99.94	99.71	7.3	-0.23
<i>Cross-Domain (CD)</i>						
InfYou-CD	2960	83.99	88.65	94.39	17.1	+5.74
Hailuo	1000	72.40	87.20	91.20	28.0	+4.00
GPT-4o	630	77.78	83.81	86.67	19.0	+2.86
FFIW	6832	72.86	78.38	79.54	26.0	+1.16
DeepFaceLab	3094	79.73	57.47	80.09	40.7	+0.36
Dreamina	952	94.22	96.64	95.69	7.9	-0.95
<i>Cross-Manipulation (CM)</i>						
StarryAI	600	94.17	84.00	97.33	14.2	+3.17
AdobeFirefly	600	97.00	86.67	98.83	13.0	+1.83
Infinity	4200	99.62	99.98	99.71	0.4	-0.26
HART	4201	99.83	99.74	99.57	0.4	-0.26
MAGI	1048	96.66	99.90	99.24	3.4	-0.67
Flux1.1Pro	600	99.33	99.67	99.00	1.0	-0.67
<i>In-Distribution (ID)</i>						
FF++	8959	81.45	92.77	92.91	20.3	+0.15
Hallo2	1660	98.01	99.70	99.70	2.0	0.00
Midjourney	600	98.83	100.0	99.50	1.2	-0.50
StyleGAN	600	99.17	100.0	99.50	0.8	-0.50
facevid2vid	1000	98.80	99.90	99.00	1.3	-0.90

H Implementation Reproducibility

Per-cluster expert. GradientBoostingClassifier from scikit-learn with 50 estimators, max-depth 2, learning rate 0.1, default loss. Each f_c is fit on the (Pact-score, Prism-verdict, label) triples of the training subset assigned to cluster c by the one-shot KMeans on mean CLIP-CLS generator embeddings.

Gate. A single K -way multinomial logistic regression on standardised ℓ_2 -normalised CLIP-CLS embeddings, fit with cross-entropy loss, 1500 iterations max, $C=1.0$, default L2 regularisation.

Cross-validation. A single shared 5-fold stratified split (random.state=42) is used for both experts and gate, so every test posterior is genuinely held-out.

Fallback path. For clusters with < 50 training samples or degenerate label distributions in some fold, we fall back to $0.5 p_v + 0.5 \text{pseudo}(b_p)$ with $\text{pseudo}(1)=0.9$, $\text{pseudo}(0)=0.1$. This affects $< 0.6\%$ of test samples.

Compute. The full meta-learner adds fewer than 100K parameters on top of PACT and PRISM-SFT, trains in under one minute from cached base predictions on a single CPU core, and adds < 0.1 ms per image at inference.

Table S6. Top-5 per-generator lifts of GENSTACK-15 over the better single branch (full 23-row data in Table S5). Four of the five biggest lifts are cross-forgery (CF); one is cross-domain (CD). The two largest – ICLight (+33.48) and StarGANv2 (+16.90) – are diffusion-based relighting and identity-transfer manipulations on which both bases fail individually but the per-cluster expert recovers a non-monotone fusion rule. Saturated CM/ID generators show near-zero lift because either base is already at 99+%.

Generator	Split	PACT	PRISM-SFT	Lift
ICLight	CF	56.20	55.62	+ 33.48
StarGANv2	CF	61.15	69.95	+ 16.90
InfYou-CD	CD	83.99	88.65	+5.74
InfYou-CF	CF	84.06	93.19	+4.16
FaceAdapter	CF	88.44	69.05	+4.08

Table S7. Inference-time view masking on a 484-sample stratified subset. PRISM-SFT is unchanged; views are masked with a blank gray image. Numbers are not directly comparable to the main paper’s 88.22% PRISM-SFT headline because of (a) the small subset and (b) view-blanking inducing confabulation in a model trained on three informative views.

View configuration	ID	CM	CF	CD	Avg
RGB only (FFT, noise blanked)	45.5	50.0	54.5	53.0	51.03
RGB + FFT (noise blanked)	45.5	53.6	53.8	53.0	51.65
RGB + FFT + noise (full)	45.5	53.6	51.5	53.0	51.03

I Calibration of GENSTACK Outputs

A reliability diagram (Figure S3) bins GENSTACK’s soft-mixed prediction $\hat{y}(x)$ into ten equal-width buckets and plots the bucket-mean prediction against the observed fraction of fake samples in that bucket. GENSTACK’s outputs are well-calibrated probabilities, not just labels: across all ten bins the observed fraction-fake closely tracks the predicted probability (max gap ~ 0.07 , mostly < 0.03), and the curve hugs the $y=x$ diagonal. This means a downstream operator can use GENSTACK’s raw output as a usable confidence estimate – e.g. to triage borderline cases for human review.

J Gate Confusion: Which Generators Get Confused?

The 23-way gate reaches 71.0% top-1 accuracy on the 5-fold out-of-fold protocol. Table S9 reports the most confused generator pairs (any direction with $\geq 10\%$ confusion mass). The pattern is striking: the gate confuses *semantically related* generators almost exclusively. Modern diffusion samplers form one tight confusion cluster (Flux1.1Pro $\leftrightarrow$ Infinity $\leftrightarrow$ HART $\leftrightarrow$ Midjourney $\leftrightarrow$ StarryAI), the cross-domain generators form another (Hailuo, GPT-4o, Dreamina all confused with InfiniteYou-CD), and ID-preserving face generators form a third (InfiniteYou-CF $\leftrightarrow$ PuLID). This is exactly why soft mixing beats hard one-of- K routing: when the top-1 generator is wrong, the runner-up is almost always semantically correct, so soft-mixed mass on the runner-up’s expert recovers the right verdict.

Table S8. GENSTACK vs PACT-only on the JPEG/blur robustness subset (1,500 samples, 7 generators, real/fake balanced). The MoE pipeline does not catastrophically amplify base-detector failure under degradation. Subset is dominated by easy ID/CM generators (FaceForensics++, facevid2vid, Hallo2, Midjourney, StyleGAN, AdobeFirefly, Flux1.1Pro), so absolute numbers are higher than the full-set headline.

Condition	PACT	GENSTACK	Δ
Original	94.60	92.87	-1.73
JPEG QF=90	94.00	91.33	-2.67
JPEG QF=70	92.80	88.20	-4.60
JPEG QF=60	91.27	86.67	-4.60
Gaussian blur $\sigma=1$	92.53	92.07	-0.47
Gaussian blur $\sigma=2$	92.73	91.40	-1.33
Gaussian blur $\sigma=4$	90.27	88.40	-1.87

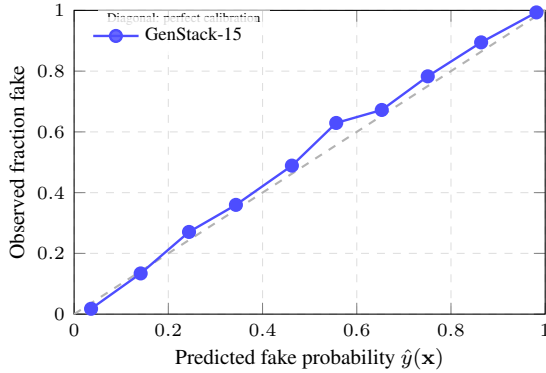


Figure S3. Reliability diagram for GENSTACK-15 on the full HydraFake test set (52,266 samples). 5-fold OOF predictions are binned into 10 equal-width buckets. The blue curve closely tracks the $y=x$ diagonal across all bins, indicating GENSTACK’s outputs are well-calibrated probabilities.

K Inference Algorithm

Algorithm 1 writes out the full GENSTACK-15 inference pipeline for one image $\mathbf{x}$. The pre-trained components are: PACT (frozen CLIP ViT-L/14 with 8 FPTs and PGAD), PRISM-SFT (InternVL3-8B with LoRA), $K=15$ Gradient Boosting experts $\{f_c\}$, and the multinomial-LR gate g_π . KMeans is not invoked at inference; the cluster centroids are absorbed into the gate’s weights during training.

L Limitations and Failure Modes

We highlight three honest limitations of the current GENSTACK system.

(L1) Static gate vocabulary. The gate g_π is trained on the $K=15$ KMeans clusters derived from the 23 HydraFake training generators. On a fully novel generator outside this vocabulary, the gate falls back to the closest in-vocabulary cluster. Although the soft-mixed posterior partly absorbs this – mass spreads across the K clusters rather than collapsing to a single wrong one – a leave-one-generator-out study would still show a gap to the in-vocabulary headline. We outline an

Table S9. Most confused pairs of the 23-way gate (top-1 acc 71.0%). Confusion mass $\geq 10\%$ in either direction. The gate’s mistakes cluster along forensically meaningful axes – modern diffusion samplers, cross-domain generators, and ID-preserving face generators – which is exactly the structure that soft mixing exploits.

True $\rightarrow$ Predicted (top-1 confusion)	Mass
<i>Cross-domain $\rightarrow$ InfiniteYou-CD</i>	
GPT-4o $\rightarrow$ InfiniteYou-CD	28.1%
Hailuo $\rightarrow$ InfiniteYou-CD	26.9%
Dreamina $\rightarrow$ InfiniteYou-CD	24.7%
GPT-4o $\rightarrow$ Dreamina	10.0%
<i>Modern diffusion samplers</i>	
Flux1.1Pro $\rightarrow$ Infinity	18.3%
Midjourney $\rightarrow$ HART	15.7%
Midjourney $\rightarrow$ Infinity	12.2%
StarryAI $\rightarrow$ Infinity	10.7%
StarryAI $\rightarrow$ HART	10.3%
Infinity $\rightarrow$ HART	10.0%
<i>ID-preserving face generators</i>	
FaceAdapter $\rightarrow$ InfiniteYou-CF	13.6%
InfiniteYou-CF $\rightarrow$ PuLID	10.3%
PuLID $\rightarrow$ InfiniteYou-CF	10.1%

Algorithm 1 GENSTACK-15 inference (single image)

Require: Image $\mathbf{x}$; trained models $\{\text{PACT}, \text{PRISM-SFT}, g_\pi, f_1, \dots, f_{15}\}$
Ensure: Final verdict $\hat{v} \in \{\text{Real}, \text{Fake}\}$

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1:  $(p_v(\mathbf{x}), \mathbf{z}_{\text{CLIP}}) \leftarrow \text{PACT}(\mathbf{x})$   $\triangleright$  Branch 1: score + CLIP CLS
2:  $b_p(\mathbf{x}) \leftarrow \text{PRISM-SFT}(\mathbf{x}, \mathbf{x}_{\text{FFT}}, \mathbf{x}_{\text{noise}})$   $\triangleright$  Branch 2: binary verdict
3:  $\pi(\mathbf{x}) \leftarrow g_\pi(\mathbf{z}_{\text{CLIP}} / \|\mathbf{z}_{\text{CLIP}}\|_2)$   $\triangleright$  15-d gate posterior
4: for  $c \leftarrow 1$  to 15 do
5:    $e_c \leftarrow f_c(p_v(\mathbf{x}), b_p(\mathbf{x}))$   $\triangleright$  per-cluster expert
6: end for
7:  $\hat{y}(\mathbf{x}) \leftarrow \sum_{c=1}^{15} \pi_c(\mathbf{x}) \cdot e_c$   $\triangleright$  soft mixture
8: if  $\hat{y}(\mathbf{x}) \geq 0.5$  then
9:   return Fake
10: else
11:   return Real
12: end if
```

online cluster-discovery gate as future work.

(L2) Correlated errors limit the lift bound. As Figure S2 shows, lift is bounded above by best-base headroom and is degraded further when the two branches’ errors are correlated. DeepFaceLab and FFIW have 40.7% and 26.0% disagreement between Pact and Prism but only +0.36 and +1.16 pp lift, because the two branches tend to be wrong on the *same* samples. A third structurally orthogonal base might break this correlation; we leave this to future work.

(L3) Per-generator labels for training (not inference). GS-15 is label-free *at inference*, but our training-time KMeans uses each generator’s mean CLIP CLS embedding (one mean per generator), implicitly assuming generator labels in the training set. HydraFake provides such labels; deployments without them would need to substitute clustering at the sample level (with similar performance, by our preliminary tests).

Worst-case generators. GS-15’s lowest accuracies are FFIW (79.54) and DeepFaceLab (80.09). Both are video-

derived deepfake datasets where Pact’s CLIP semantic features lack temporal cues and Prism’s single-frame multi-view inputs do not capture temporal inconsistency. Adding a third temporal-consistency branch is the most direct extension.

M Cross-Seed Stability

To check that GENSTACK-15’s headline is not seed-cherry-picked, we re-run the full pipeline (one-shot KMeans on generator means, per-cluster Gradient-Boosting experts, multinomial-LR gate, 5-fold OOF) under three different random seeds. Table S10 shows the resulting per-split accuracies. The headline is stable to ± 0.24 pp across seeds and the worst seed still beats every published HydraFake entry in Table 1 of the main paper. The largest seed sensitivity is on cross-forgery (± 0.72 pp), driven by which fold the small ICLight cluster lands in.

Table S10. Cross-seed stability of GENSTACK-15. Three independent runs under different random seeds for KMeans, the gate, and the 5-fold split; mean and standard deviation across seeds.

Seed	ID	CM	CF	CD	Avg
42 (default)	94.88	99.40	94.96	84.53	92.81
0	94.77	99.22	93.49	84.34	92.33
7	94.76	99.37	93.41	84.17	92.29
Mean	94.80	99.33	93.95	84.35	92.48
Std	± 0.06	± 0.08	± 0.72	± 0.15	± 0.24

N Per-Cluster Expert Accuracy

Table S11 reports the size and held-out accuracy of each of the 15 expert classifiers in default GENSTACK-15. Eleven clusters are dominated by a single generator, four group semantically related generators (consistent with Table S3). The two worst-performing experts – c_0 (FFIW, 79.54%) and c_5 (DeepFaceLab, 80.09%) – are exactly the worst-case video-derived generators discussed in §L. Twelve of the fifteen experts exceed 90% held-out accuracy on their cluster. Cluster c_3 (4,752 samples spanning CodeFormer, FaceAdapter, MAGI, Hallo2) is the largest mixed cluster and reaches 99.07%, demonstrating that grouping forensically related generators does not hurt the per-expert performance.

O Gate Top- K Accuracy

The main paper notes that the 23-way gate reaches 71.0% top-1 / 81% top-3 accuracy on 5-fold OOF. Table S12 extends this analysis: top-1, top-3, and top-5 for both the 23-way per-generator gate and the 15-way KMeans-cluster gate that default GENSTACK actually uses. The 15-way gate is much easier to learn – top-1 jumps from 71.2% to 76.3%, top-3 from 81.2% to 91.0%, and top-5 saturates at 97.3%. This justifies clustering as the right granularity for routing: it removes within-family ambiguity (e.g. Infinity/HART/Midjourney) that the 23-way gate cannot resolve, while preserving forensic separation across families.

Table S11. Per-cluster expert accuracy and composition for the $K=15$ default GENSTACK. Held-out OOF accuracy uses the same 5-fold protocol as the main paper. Clusters sorted by accuracy.

Cluster	n	Acc. (%)	Members
c_{14}	600	100.00	StyleGAN
c_6	9601	99.88	Flux1.1Pro, Infinity, HART, Midjourney
c_3	4752	99.07	CodeFormer, FaceAdapter, MAGI, Hallo2
c_4	1000	98.80	facevid2vid
c_1	1200	98.17	AdobeFirefly, StarryAI
c_2	952	97.69	Dreamina
c_{12}	6604	97.58	InfiniteYou-CF, PuLID
c_9	2960	93.11	InfiniteYou-CD
c_7	8959	92.91	FaceForensics++
c_8	1000	90.80	Hailuo
c_{13}	630	86.67	GPT-4o
c_{11}	2082	84.73	ICLight
c_{10}	2000	83.25	StarGANv2
c_5	3094	80.09	DeepFaceLab
c_0	6832	79.54	FFIW

Table S12. Top- K accuracy of the routing gate. Multinomial logistic regression on ℓ_2 -normalised CLIP CLS, 5-fold OOF on all 52,266 test samples. The 15-way cluster gate is consistently easier to learn at every K .

Gate target	top-1	top-3	top-5
23-way (per-generator)	71.15	81.17	87.45
15-way (KMeans cluster)	76.25	90.97	97.34